Thompson's Group F, Quantum Coherence, and the Photoelectric Effect

A Dyadic-Symmetry Correspondence Hypothesis and Testable Research Program

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Abstract

Can representations of Thompson's group F describe optical transformations and their photoelectric readouts? We approach this question through dyadic interval maps, normalized states and detector probabilities. The mathematical starting point is the group of increasing dyadic piecewise-linear homeomorphisms of the unit interval. OpenAI's September 2026 preprint reports its nonamenability [1,2]. Earlier work connects Thompson-group representations with quantum models and field-like observables [3–7].

We use two constructions. A weighted-composition representation on $L^2([0,1])$ describes changes in spatial detection probabilities. The left-regular representation on $\ell^2(F)$ describes state displacement through the identity

$$\boxed{\|\lambda(s)|\psi_A\rangle - |\psi_A\rangle\|^2 = \frac{|sA \triangle A|}{|A|}}.$$

The identity relates a finite-set boundary ratio to the distance between normalized Hilbert-space vectors. With a specified optical encoding, the same overlap determines ideal interferometric visibility. We then introduce a separate detector response and the photoelectric threshold $K_{\max} = h\nu - \Phi$. The proposed relation is descriptive: the representation of F organizes the optical transformations, while the detector model supplies the light–matter response. Its physical implementation remains to be tested.

Keywords: Thompson's group F; nonamenability; photoelectric effect; quantum optics; dyadic partitions; interferometry; coherence; photon detection; Hilbert-space representations; Følner condition.

1 Scope and research question

We ask whether a representation of F can describe the redistribution of optical coherence and the resulting photoelectric detection statistics. To make the question concrete, we specify the states, transformations and measured probabilities. We also examine how the finite-set obstruction associated with nonamenability appears in one particular representation.

The investigation uses three bodies of work:

Basis	Role in the model	Sources
Group theory	Dyadic transformations, representations and the reported nonamenability result	[1,2]
Quantum mathematics	Unitary representations, isometric splitting and field-like correlations	[3–7]
Optical and photoelectric physics	Energy thresholds, detector response and optical correlations	[8–11]

We begin with the mathematical structure, introduce the optical quantities and then join the two through explicit probability models. The proposed experiments test those models under stated assumptions.

2 Thompson's group F and its mathematical meaning

Let

$$D = \{\frac{k}{2^n} : k \in \mathbb{Z},\ n \in \mathbb{N}_0\} \cap [0,1].$$

The group F consists of increasing homeomorphisms $f : [0,1] \to [0,1]$ having finitely many affine pieces, breakpoints in D, and slopes 2^m, $m \in \mathbb{Z}$, wherever the derivative exists. Its operation is function composition $fg = f \circ g$.

Dyadic subdivision provides a hierarchy of interval scales. Each group element is reversible: it is a continuous bijection with a continuous inverse. Absorption and the electronic detector record enter through the separate response model in Section 7.

A concrete element is

$$f_0(x) = \begin{cases} x/2, & 0 \le x \le 1/2, \\ x - 1/4, & 1/2 \le x \le 3/4, \\ 2x - 1, & 3/4 \le x \le 1. \end{cases} \tag{1}$$

Its breakpoints are $1/2, 3/4$, its slopes $1/2, 1, 2$, and it maps the respective input intervals onto $[0, 1/4]$, $[1/4, 1/2]$, $[1/2, 1]$.

2.1 Amenability and finite-set displacement

For a discrete group G, a positive normalized invariant mean is a functional

$$\begin{aligned} M : \ell^\infty(G) &\longrightarrow \mathbb{R}, \qquad M(1) = 1, \\ f \ge 0 &\Longrightarrow M(f) \ge 0, \\ M\big(g \mapsto \varphi(sg)\big) &= M(\varphi). \end{aligned} \tag{2}$$

The Følner criterion says amenability entails, for each finite $S \subset G$ and every $\varepsilon > 0$, some nonempty finite $A \subset G$ satisfying

$$\frac{|sA \triangle A|}{|A|} < \varepsilon \quad (s \in S). \tag{3}$$

Here $\triangle$ denotes symmetric set difference. The numerator counts elements belonging to exactly one of sA and A. Nonamenability is equivalent to a uniform obstruction of the following form, used here with the reported result for F [1]:

$$\begin{aligned} &\exists\, S \Subset F, \quad \exists\, b > 0 : \\ &\forall\, \varnothing \ne A \Subset F, \quad \exists\, s \in S : \quad \frac{|sA \triangle A|}{|A|} \ge b. \end{aligned} \tag{4}$$

Here $A \Subset F$ means that A is a finite subset of F. The original preprint builds recursive Hilbert-space-valued "colors" of dyadic partitions and compares scalar correlations. It also constructs a Lipschitz map on an infinite-dimensional Hilbert ball with a nonzero displacement. Its boundary parameter can be expressed, for an integer $D > 4L^2/\delta^2$, as

$$b(D) = \frac{\delta^2/L^2 - 4/D}{4(1 - 1/D)} > 0. \tag{5}$$

The preprint constructs a displacement witness with $\delta = 1/2$ and an associated Lipschitz bound $L > 0$. Its Hilbert-space-valued "colors" are vectors used in the proof. An optical interpretation requires a separate assignment of physical modes and operations.

The OpenAI preprint is dated 23 September 2026. Kim and Lodha's note, submitted on 13 October 2026, provides a human-written exposition [2]. The Lean scope document states

nonamenability without specifying a boundary constant or a prescribed generating set [1]. We use the reported theorem as a premise; we have not independently rebuilt its formalization.

3 Representations and quantum models

Vaughan F. R. Jones constructed unitary representations of F, T using ideas motivated by quantum-field theory and local changes of scale [3]. Brothier and Jones developed Pythagorean representations from operators A, B on a Hilbert space $\mathcal{K}$ satisfying

$$\boxed{A^\dagger A + B^\dagger B = I_{\mathcal{K}}}. \tag{6}$$

The map $V : \mathcal{K} \to \mathcal{K} \oplus \mathcal{K}$, defined by $V\xi = (A\xi, B\xi)$, preserves norm:

$$\|A\xi\|^2 + \|B\xi\|^2 = \|\xi\|^2. \tag{7}$$

These constructions are documented in [4] and extended in [7]. Osborne and Stiegemann introduced quantum-field-like observables in a semicontinuous limit of such representations [5], and Stiegemann examined related conformal-field-theory toy models [6].

Set $E_0 = A^\dagger A$ and $E_1 = B^\dagger B$. These are positive operators with $E_0 + E_1 = I$, so they also define a complete two-outcome POVM. This common normalization gives a way to compare isometric splitting with quantum measurement. A detector interpretation still requires a physical model for the operators.

4 Optical quantities and detector response

4.1 Propagation and optical frequency

Water-wave comparisons illustrate wavelength and interference. An electromagnetic description also needs frequency, phase and field correlations. These quantities let us state the proposed correspondence directly.

For a vacuum monochromatic wave,

$$\omega = 2\pi\nu, \qquad k = 2\pi/\lambda, \qquad c = \omega/k = \lambda\nu. \tag{8}$$

For a photon,

$$E_\gamma = \hbar\omega = h\nu, \qquad p_\gamma = \frac{h}{\lambda}. \tag{9}$$

A photon has no rest frame. In a medium, the phase velocity is $v_{\mathrm{ph}} = \omega/k$, often written as $c/n(\omega)$, while the group velocity is $v_{\mathrm{g}} = d\omega/dk$. At an interface between stationary media, frequency is conserved and the wave vector changes according to the dispersion relation. Phase velocity, group velocity, absorption and loss of coherence describe different aspects of propagation.

4.2 Photoelectric threshold

For idealized external photoemission in the single-photon regime,

$$\begin{aligned} K_{\max} &= h\nu - \Phi, \\ \nu_0 &= \frac{\Phi}{h}, \qquad h\nu \geq \Phi. \end{aligned} \tag{10}$$

Here Φ is the material work function and ν_0 is the threshold frequency [8]. Real measurements also depend on band structure, the surface, temperature and the instrument. In CCD/CMOS devices, absorbed light can generate an electronic excitation that is subsequently registered. This internal detector response differs from external electron emission.

4.3 Detection probabilities and coherence

Glauber's theory describes optical coherence through field correlations, including

$$G^{(1)}(x,t;x',t') = \left\langle \hat{E}^{(-)}(x,t)\hat{E}^{(+)}(x',t') \right\rangle . \tag{11}$$

Normally ordered higher-order functions govern coincidence probabilities [9]. More generally, for a state ρ and a positive-operator-valued measure (POVM),

$$\begin{aligned} P(j \mid \rho) &= \mathrm{Tr}(\rho E_j), \\ E_j \succeq 0, \qquad & \sum_j E_j = I. \end{aligned} \tag{12}$$

If a unitary representation $U(g)$ acts before detection,

$$P(j|g,\rho) = \mathrm{Tr}\Big(U(g)\rho U(g)^\dagger E_j\Big) . \tag{13}$$

The detector outcome is a physical measurement record.
For two optical paths, take

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{14}$$

Which-path information can become correlated with the environment:

$$|\Psi\rangle = \alpha|0\rangle|e_0\rangle + \beta|1\rangle|e_1\rangle. \tag{15}$$

Tracing over the environment yields

$$\rho_{01} = \alpha\beta^*\langle e_1|e_0\rangle. \tag{16}$$

Orthogonal environmental records make the off-diagonal term vanish in this basis [10,11]. The combined system can remain pure while the reduced optical state loses coherence. Absorption requires its own light–matter interaction model.

4.4 The "fragile fabric" and "quiescent gas" analogies

The fragile-fabric analogy focuses on relationships: a visible pattern depends on how its parts are connected, and a local disturbance can change that pattern. For optical coherence, Eq. (16) gives a precise counterpart: the interference term depends on the overlap between environmental states. The analogy concerns correlations; it does not assign material threads to photons.

A gas distribution offers a second image for the effect of a local interaction on collective behavior. It motivates a question about sensitivity and statistical description. Any prediction about optical coherence or entropy must then follow from the specified state and dynamics.

5 Mathematical model I: spatial optical modes

Take $\mathcal{H}_x = L^2([0,1], dx; \mathbb{C})$. For $f \in F$, define the weighted-composition operator

$$\boxed{(U_f\psi)(x) = \sqrt{(f^{-1})'(x)}\,\psi(f^{-1}(x))} \quad \text{almost everywhere.} \tag{17}$$

A change of variables proves unitarity and the representation property:

$$\int_0^1 |U_f\psi(x)|^2 dx = \int_0^1 |\psi(y)|^2 dy, \qquad U_f U_g = U_{f\circ g}. \tag{18}$$

For an ideal spatial detection bin I,

$$\begin{aligned} P(I \mid f, \psi) &= \int_I |U_f \psi(x)|^2 \, dx \\ &= \int_{f^{-1}(I)} |\psi(y)|^2 \, dy. \end{aligned} \tag{19}$$

Worked dyadic example. Choose the f_0 of Eq. (1) and normalized uniform input $\psi(x) = 1$. For bins $I_1 = [0, 1/4]$, $I_2 = [1/4, 1/2]$, $I_3 = [1/2, 1]$,

$$|U_{f_0}\psi(x)|^2 = \begin{cases} 2, & x \in I_1, \\ 1, & x \in I_2, \\ 1/2, & x \in I_3. \end{cases} \tag{20}$$

The corresponding probabilities change from $(1/4, 1/4, 1/2)$ before the transformation to

$$\boxed{(P_1, P_2, P_3) = (1/2, 1/4, 1/4)} \tag{21}$$

afterward. The probabilities still sum to one. This example describes spatial redistribution under a specified unitary map; a device implementing that map can be tested against these bin weights.

6 Mathematical model II: nonamenability and interferometric state displacement

Consider the left-regular representation of F on $\ell^2(F)$, whose orthonormal basis consists of vectors $|g\rangle$, $g \in F$. Define

$$\begin{aligned} \lambda(s)|g\rangle &= |sg\rangle, \\ |\psi_A\rangle &= \frac{1}{\sqrt{|A|}} \sum_{g \in A} |g\rangle. \end{aligned} \tag{22}$$

for every finite nonempty $A \subset F$.

Proposition 1 (finite-support displacement identity). For each $s \in F$,

$$\langle \psi_A | \lambda(s) | \psi_A \rangle = \frac{|A \cap sA|}{|A|} \tag{23}$$

and

$$\boxed{\| \lambda(s)|\psi_A\rangle - |\psi_A\rangle \|^2 = \frac{|sA \triangle A|}{|A|}}. \tag{24}$$

Derivation. Orthonormality gives Eq. (23). Since both vectors have unit norm, the squared distance equals $2 - 2|A \cap sA|/|A|$. The equality $|sA| = |A|$ implies $|sA \triangle A| = 2|A| - 2|A \cap sA|$; substitution yields Eq. (24).

Equation (24) holds for every discrete group in its left-regular representation. The role of nonamenability enters through the quantifiers in Eq. (4): one fixed finite set S and one positive bound must work for every finite nonempty A. Together, Eqs. (4) and (24) imply

$$\forall \varnothing \neq A \Subset F, \ \exists s \in S : \quad \| \lambda(s)|\psi_A\rangle - |\psi_A\rangle \| \geq \sqrt{b} \tag{25}$$

and

$$\frac{|A \cap sA|}{|A|} \leq 1 - \frac{b}{2}. \tag{26}$$

Conditional optical reading. If an apparatus can prepare mutually orthogonal modes $|g\rangle$ and coherently apply $\lambda(s)$, a balanced two-arm interferometer comparing $|\psi_A\rangle$ with $\lambda(s)|\psi_A\rangle$ ideally gives fringe visibility

$$\begin{aligned} V_{s,A} &= \left|\langle\psi_A|\lambda(s)|\psi_A\rangle\right| \\ &= \frac{|A \cap sA|}{|A|}. \end{aligned} \tag{27}$$

Thus at least one element of the fixed finite set S would satisfy

$$\boxed{V_{s,A} \leq 1 - b/2} \tag{28}$$

for each equal-phase finite-support state of Eq. (22), with s allowed to depend on A. The restriction to this state family matters. A finite optical implementation tests its encoded states and transformations; it does not establish a statement over all finite subsets of the infinite group.

The same overlap can be measured with coherent classical fields. Comparing classical-field and single-photon implementations is therefore part of the proposed verification program.

6.1 Amenable negative control

The same identity applies to the amenable additive group $\mathbb{Z}$. With $A_N = \{-N, \ldots, N\}$ and shift $s = 1$,

$$\frac{|(A_N + 1)\triangle A_N|}{|A_N|} = \frac{2}{2N+1} \longrightarrow 0. \tag{29}$$

Here the boundary ratio approaches zero as the support grows. The comparison shows why a nonzero displacement in one experiment does not distinguish amenability. The relevant question is whether a fixed finite family enforces a positive bound for every finite support.

7 Joining optical probability distributions to photoelectric readouts

The Pythagorean splitting in Eqs. (6)–(7), the coordinate map in Eqs. (17)–(19), and a detector POVM are mathematically compatible, but serve different roles.

For an optical mode ψ, transformed by U_f, with detector bin I_j and photon-frequency-dependent efficiency $\eta_j(\nu)$, an idealized single-photon model is

$$P(\text{click in } I_j \mid f, \nu, \psi) = \eta_j(\nu) \int_{I_j} |U_f\psi(x)|^2 \, dx. \tag{30}$$

Assume $0 \leq \eta_j(\nu) \leq 1$ and bins that partition the coordinate interval. Adding a no-click outcome normalizes the probabilities. For an idealized external photoelectric detector, set $\eta_j(\nu) = 0$ when $h\nu < \Phi$. Above threshold, the response depends on the material and device. The integral describes where an excitation is registered; the response function describes its detection efficiency, and Eq. (10) gives the maximum emitted-electron energy in the idealized regime.

Together, these components describe a photoelectric readout for a specified optical state. The representation of F supplies the spatial transformation; the detector response supplies absorption and registration. The work function is an input from the material model. A microscopic extension would need a light–matter Hamiltonian and the dynamics of the electronic record.

8 Structural terms and optical interpretation

Mathematical expression	Structure it describes	Proposed optical role
Dyadic piecewise	Binary rational subintervals	A hierarchy of spatial bins
Increasing piecewise-linear homeomorphisms	Reversible interval maps	Unitary transformations of optical modes
Nonamenability	Failure of the Følner approximation	A conditional displacement bound in the left-regular representation
Recursive colors and Hilbert correlations	Vectors and inner products in the proof	A prompt to specify physical modes and measurable correlations

The table identifies the level of each correspondence. A physical implementation must specify how a mode, a group action and a measured outcome are related. Sections 5–7 give candidate descriptions that can be checked mathematically and tested in an optical setup.

9 Proposed verification program

Phase A — mathematical reproduction. Check Eqs. (17)–(29) analytically and computationally. State which operations implement the group action and which are finite approximations. Review the formalization and the mathematical literature. A numerical test of the uniform bound also needs a concrete set S and a justified value of b; these are not supplied by the published Lean scope statement [1].

Phase B — optical implementation. Encode normalized amplitudes in dyadic transverse modes and approximate selected maps U_f with calibrated optical networks. Measure detection-bin weights and interference overlap using both coherent classical fields and single-photon sources. Report optical loss, mode matching, detector efficiency, statistical uncertainty and calibration.

Phase C — detector response and physical specificity. Couple the optical description to absorption and electronic degrees of freedom. Check energy conservation, the material work function and the regime in which $K_{\max} = h\nu - \Phi$ applies. Compare the predicted counts and emitted-electron energies with conventional photodetection theory. A claim beyond a descriptive correspondence would require an additional quantitative prediction, a control that distinguishes it from the conventional model and conditions under which it fails.

This program can test whether the representation describes the chosen setup and whether it offers a useful way to organize its measurements. Any additional physical claim depends on the outcome of that comparison and independent replication.

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The shared dialogue [12] records conceptual provenance. It is not used as evidence for a mathematical or physical result. Elementary consistency checks accompany the manuscript in the repository.

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